Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_31

1. Summary of Key Deviations

The analysis of Run_31 reveals a minor, but physically contradictory, set of symptoms that point to a specific component malfunction.

- **Flow_Bypass (Moderate):** The most significant anomaly is a moderate **negative offset**, with bypass flow reading **-10% to -18%** lower than healthy. This is a key diagnostic clue, as it's the inverse of a typical blockage fault.
- **Flow_Pump (Minor):** A corresponding minor **negative offset** of around **-3% to -4%** is seen on the pump flow.
- **Flow_Engine1-4 (Minor):** In direct contradiction to the negative pump flow, all four engine flows show a minor **positive offset** of **+3% to +7%** during the high-throttle phases.
- **Pressure Signals (Nominal):** All pressure sensors across the system (`p_Pump`, `p_FMU`, `p_Combustion`, `p_Shut`) show only negligible deviations, indicating no significant pressure changes.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Bypass Valve**	**Stuck Partially Closed / High Cracking Pressure**	This is the most compelling hypothesis. A bypass valve that is stuck partially closed or has a higher-than-normal cracking pressure would fail to open correctly to divert flow. This would force more fuel than intended downstream towards the engine, perfectly explaining why the **Flow_Engine** signals are all positive. This increased downstream flow and resistance would cause the overall pump flow (`Flow_Pump`) to drop slightly, and because the valve isn't opening as much, the measured `Flow_Bypass` would also be lower than normal. This single fault explains the full set of contradictory symptoms.

2	**Flow_Bypass & Flow_Engine Sensors**	**Dual Sensor Calibration Error**	It's possible the `Flow_Bypass` sensor has a negative offset and the `Flow_Engine` sensors all have a positive offset. However, it's less likely for five separate sensors to all fail simultaneously in such a specific, opposing manner than for a single physical component to cause these interrelated readings.
3	**Fuel Metering Unit (FMU)**	**Over-delivering Fuel**	If the FMU were miscalibrated and delivering too much fuel, this would explain the positive `Flow_Engine` readings. However, it would not explain why both `Flow_Pump` and `Flow_Bypass` are reading *lower* than normal.
4	**Main Fuel Pump**	**Minor Degradation**	Pump degradation would explain the minor drop in `Flow_Pump` and `Flow_Bypass`. However, it absolutely cannot explain why the downstream `Flow_Engine` would *increase*. A failing pump should lead to lower, not higher, engine flow.
5	**System Contamination**	**Debris causing valve to stick**	This is the likely underlying cause of the stuck valve (Hypothesis 1). A piece of debris could be preventing the bypass valve from opening correctly.

3. Most Probable Hypothesis

The most probable root cause is that the **Bypass Valve is Stuck Partially Closed or has a high cracking pressure.**

Reasoning: This single mechanical failure is the most parsimonious explanation for the otherwise contradictory data. The core of the issue is explaining how engine flows can be high while the pump and bypass flows are low. A bypass

valve that fails to open properly does exactly this. It essentially creates a partial blockage in the bypass return loop, forcing more fluid into the downstream part of the system than the controls intend. This unwanted downstream flow is registered as a positive offset on the `Flow_Engine` sensors. In response to this higher-than-expected system resistance, the fixed-displacement pump's total output flow drops slightly (`Flow_Pump` is low), and because the valve isn't opening as it should, the measured `Flow_Bypass` is also low. This single event explains the entire symptom set.